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Cyber-susiliency index: A comprehensive resiliency-sustainability-cybersecurity index for healthcare supply chain networks

Niamat Ullah Ibne Hossain [a,*], Sazid Rahman [b], Sharmine Akther Liza [c]

[a] Engineering Management, College of Engineering and Computer Science, Arkansas State University, Jonesboro, AR, USA
[b] Industrial Engineering, University of Arkansas, Fayetteville, AR, USA
[c] Industrial and Systems Engineering, Lehigh University, Bethlehem, PA, USA

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ABSTRACT

Resilience, sustainability, and cyber-security are three measures in supply chains that define how robustly supply chains perform without disruption. Although these three measures can each be assessed separately and give us some idea of a supply chain's capability, a combined measure of all three can provide a more holistic picture and comprehensive understanding of the system. To our knowledge, no attempt has been undertaken to formulate a combined measure. We utilized a data fusion technique called the Dempster Shafer (D–S) theory and developed a comprehensive assessment framework considering all three dimensions of resilience, sustainability, and cyber-security. Further, we combine these three measures into a single measure called the cyber-susiliency index. The proposed framework is applied to a healthcare supply chain and shows reasonable results. Yager's recursive rule of combination further validates the D–S theory application. An index value is developed on a scale from 0 to 1, representing a numerical measure of the organization's cyber-susiliency. The cyber-susiliency index for the healthcare supply case was calculated at 0.62 on a 1 to 0 scale.

1. Introduction

The Healthcare supply chain (HCSCs) is a complex adaptive ecosystem that enables the cost-effective distribution of healthcare items to patients [1]. Healthcare is considered one of the critical infrastructures, especially for human presence. With the COVID-19 pandemic causing devastating loss of wealth and human life, the importance of establishing a robust healthcare system is underscored. In order to deal with any failure/disruption in healthcare, the healthcare industry, its suppliers, distributors, service providers, and customers act together to ensure best practices are implemented [2].

When assessing the level of operational complexity and contemporary challenges pertaining to Supply Chain (SC) networks, perhaps HCSC is considered one of the most vulnerable supply chain networks which are susceptible to different types of disruptions compared to any other SC networks [3]. Due to the risks and challenges presented by the COVID-19 pandemic, the regular operations of HCSCs have become even more critical [4]. For instance, the global market has confronted a sudden surge of demand in healthcare items due to stockpiling and panic buying of pharmaceuticals and medical supplies such as face shields, masks, and sanitizers, which leads to the ultimate disruptions [5]. In the midst of the COVID-19 pandemic, the crisis grew more complex as increased demand for healthcare items could not be met. To deal with such disruptions in HCSC, there is an obvious need to establish a resilient and sustainable SC structure that will withstand the catastrophes of disruptions and eventually lessen its consequences [6]. Along with resilience, HCSCs need to consider sustainable practices as a substantial concern because of their importance in improving healthcare services in several dimensions [7]. According to a report on American healthcare, healthcare spending in the United States saw a 9.7% increase ($4.1 trillion) in 2020 compared to the 4.3% increase in 2019. This acceleration was caused by a 36% increase in federal healthcare spending due to the COVID-19 pandemic [8]. On the other hand, the cloud security firm Bitglass reported an increase of 55% in cyberattacks against U.S. healthcare organizations. According to Cybersecurity Ventures, there is a plan taken by the healthcare industry to spend around $125 billion to strengthen their cybersecurity by 2025 [9]. Although considerable forces act behind this spending, unprecedented response towards the COVID-19 pandemic strikes the most, reaffirming the significance of a resilient, sustainable and cybersafe HCSCs framework. Furthermore, although future healthcare expenditures are still hypothetical, it is evident that investments will increase for widespread vaccination programs and the possible emergence of new variants in the future. All the above discussion boils down to the point that the existing HCSC requires a resilient, sustainable, and cyber

* Corresponding author.
E-mail addresses: nibnehossain@astate.edu (N.U.I. Hossain), mdsazidr@uark.edu (S. Rahman), shl622@lehigh.edu (S.A. Liza).

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secure network that can withstand disruptions and uncertainty and run seamlessly.

Sustainability and resilience are emerging requirements in today's healthcare supply chains. Due to the shared emphasis on overall survivorship, sustainability and resilience share common study methodologies such as life-cycle analysis, structural analysis, and socioeconomic analysis of healthcare supply chain and used interchangeably in a number of contexts [10]. In addition to international frameworks and multilateral agendas that support sustainability and resilience in urbanization, resilience and sustainability are also connected to global political trends. These overlapping objectives and use cases have caused some government agencies, healthcare's, academic institutions, and charitable organizations to consolidate their sustainability and resilience departments. Despite their close relationship, these two concepts are distinct. In Marchese et al. [11], three frameworks are presented that describe the relationship between sustainability and resilience. The concept of resilience is part of the larger concept of sustainability, according to the first viewpoint. This perspective assumes that greater resilience leads to greater sustainability, but not vice versa. The second takes the opposite standpoint with sustainability as a subset of resilience. This viewpoint emphasizes the impact of sustainability on resilience. The third perspective discards a hierarchical structure between sustainability and resilience and instead suggests that resilience and sustainability are different, but when combined, augment the overall performance of a system. This perspective of an overlap or intersection between sustainability and resilience is also shared by other scholars [12–15]. For example, Ivanov [12] notes that the design of global supply chains to reduce costs has an impact on the use and consumption of natural resources, employment, and disruptions. It is also an appropriate way to think about the interrelationship between the two concepts as the agenda driving sustainable and resilient supply chains strengthens.

With these emerging risks in healthcare supply chains, resilience and sustainability must be treated together as integral parts of a robust healthcare supply chain. As a way of capturing and combining these two concepts, this paper coined the term "*Susiliency*" to describe a supplier's overall robustness in terms of sustainability and resilience. By considering resilience and sustainability together, the intent is to facilitate a more comprehensive and definitive decision-making process regarding robust healthcare supply chain networks.

Being said that there is a lack of research on evaluating *Susilient* competency while considering the likelihood of cyber-attack in the HCSC, especially when using multiple criterion decision analysis methodologies that take subjective uncertainty into account. Uncertainty resulting from incomplete information or inconsistency in observation is referred to as subjective uncertainty [16]. This ambiguity is consistent with the circumstances of subjective evaluation in healthcare. In order to address this, the Dempster–Shafer (DS) Theory can be applied to aggregate data from many sources and produce a comprehensive assessment of the competency of the healthcare supply chain. The study looks at competency traits including cybersecurity, sustainability, and resilience. This study also intends to provide specific answers to questions about cyber susiliency evaluation index in the healthcare supply chain by introducing a novel assessment framework.

With this discussion, this research was undertaken to attempt to answer the following research questions (RQs).

RQ1: How to combine and comprehend three different measures (resilience, sustainability, and cyber-security) of a *supply chain network especially for healthcare and assess the integrate state numerically as a whole?*

RQ 2: How the identified measures Resilience, sustainability, and cybersecurity for can be structured to know their importance?

RQ2: How to develop a holistic method to model and aggregate the condition of selected factors to obtain a holistic performance index of the cyber saliency of healthcare supply chain?

To respond aforementioned questions, this study identifies the factors for assessing the features of HCSCs considering resilience, sustainability, and cyber-security employing Dempster–Shafer (DS) theory. The combined term of resilience, sustainability, and cyber-security is tagged as cyber-susiliency. Then, based on these research questions, the major goal of this research is to create a framework for evaluating the combined resilience, sustainability, and cybersecurity of a Healthcare Supply Chain (HCSC) using a data fusion technique. To accomplish this goal, a thorough assessment of the currently existing literature on the resilience, sustainability, and cybersecurity of HCSC will be done in order to determine the components that influence the overall capability measure of HCSC. To describe the interrelationships among the listed elements, a hierarchical model will be constructed. Based on the acquired data, the Analytic Hierarchy Process (AHP) will be employed to establish the relative importance of the components identified and assign weights. In addition, a data fusion technique will be employed for combining subjective and objective data to assess HCSC's overall *Susiliency* measure index. Finally, practitioners are expected to use the proposed framework as a decision-making tool for assessing the combined resilience, sustainability, and cyber-security of HCSC.

In a nutshell, this study presents three contributions in the context of healthcare supply chain management to answer the research questions at hand. For starters, it highlights the key measures of a susilient supply chain in terms of resilience, sustainability, and cyber-security. Second, this research suggests an integrated approach to assessing the state of cyber-susilient competency in a HCSC. Finally, the suggested framework's practical implications are presented, emphasizing its utility for HCSC management decision-making.

2. Literature review

As the significance of HCSCs are immense and has gained tremendous attention in recent days, it requires a thorough analysis of the existing framework that will assist in responding to the aroused problems more effectively. Many researchers have worked on HCSCs considering the perspectives of resilience, sustainability, and cyber security. They conducted their study through conceptual, analytical, and statistical approaches. This section summarizes the studies that have been conducted focusing on the perspectives of resilience, sustainability, and cybersecurity of HCSCs.

2.1. Resilience in HCSCs

Healthcare resilience has been considered to provide useful insights for comprehending overall healthcare systems. However, establishing resilience in HCSCs has become more challenging regarding the continuous turbulence and disruptions it confronts at each stage. Lyng et al. [17] discuss the capacities for resilience in healthcare considering existing studies and knowledge. They analyzed and synthesized the results of existing studies and validate them by employing semi-structured interviews. Finally, they developed a conceptual framework consisting of several resilience capacities that are linked with contextualization and collaboration. Alemsan et al. [18] tried to build a relationship between lean practices and resilience capabilities in HCSCs. This study reviewed existing articles and analyzed a conceptual framework for the adaption of lean practices to establish resilience capabilities in HCSCs. This will help the practitioners to prioritize the practices to enhance resilience capabilities. Apart from the conceptual framework, researchers also focused on analytical or mathematical approaches to HCSCs resilience. Ali and Kannan [19] reviewed several modeling-based literatures including risk and resilience, the Covid-19 pandemic, and sustainability in healthcare operations. They reviewed several models such as Markov, mixed-integer programming, simulation, queueing model, and stochastic in multidimensional healthcare operations. The comparison will help researchers to comprehend current developments in the specialty and employ the most effective one. Salehi et al. [20]

proposed a resilience framework for optimizing the performance of the pharmaceutical supply chain (PSCs). They formulated the problem employing data envelopment analysis (DEA) and fuzzy data envelopment analysis (FDEA) and identified redundancy as the most effective factor in enhancing the efficiency of PSCs. Moreover, Sabouhi et al. [21] also proposed an integrated hybrid approach to design resilient PSCs. They employed two approaches namely, fuzzy DEA and the possibilistic-stochastic programming model for supplier selection and determining resilience strategies. The model demonstrates the impacts of changes in parameters and effective resilience strategies as required. Singh and Parida [22] discussed a systemic literature review of several supply chain disruption models in the healthcare sector. This study analyzed different modeling approaches such as mixed-integer linear programming, system approach, hybrid, and optimization models and identified limitations considering risk-specific criteria.

Besides that, Multicriteria decision-making (MCDM) assists in determining which factors contribute to resilient HCSCs. Zamiela et al. [6] employed a rank reversal proximity index MCDM method to prioritize the enablers which are necessary for supply chain resilience and concluded *redundancy, collaboration, and robustness* as the prominent indicators of resilience. Rehman and Ali [23] prioritized resilience strategies for healthcare supply chains considering different levels of risk. This study employed a fuzzy-based AHP-Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) for risk prioritization and fuzzy quality function deployment (QFD) to identify resilience strategies against risks. The findings focused on considering multiple sourcing, risk awareness, agility, and global diversification of suppliers. Das et al. [24] incorporate Analytic Hierarchy Process (AHP) and Decision-Making Trial and Evaluation Laboratory (DEMATEL) to analyze factors that affected the supply chain resilience networks. This study concludes "cost–optimization" is the most significant factor whereas human resource management is the least influential factor in lessening vulnerabilities of the supply chain network. Kaushik et al. [25] identify the factors that cause disruption in the healthcare supply chain amidst pandemics and epidemics. This study implies Best and Worst Method (BWM) to pairwise compare the disruption factors and concluded "inefficient executive leadership" is the most crucial disruption factor among all. Khan et al. [26] proposed a resilience-based model for the HCSCs to defeat the consequences of Covid-19. This study employed the full consistency method (FUCOM) to prioritize the resilience attributes and fuzzy quality function deployment (QFD) to prioritize the resilience strategies. The findings suggest ed "leadership and governance capacity" to be the most prominent attributes whereas "building the operational capacity of the management to be the most significant strategy.

2.2. Sustainability in HCSCs

The recent Covid-19 pandemic has largely emphasized the significance of incorporating sustainability considerations in all aspects of HCSCs. Many researchers have adopted sustainability practices in HCSCs to establish a more integrated network over different horizons. Hussain et al. [27] recognized the motivators, barriers, and enablers of social sustainability in a healthcare supply chain by employing stakeholder theory and the Suppliers, Inputs, Process, Outputs and Customers (SIPOC) technique. This study develops a comprehensive social sustainability framework that incorporates the perspectives of stakeholder groups. Maghsoudi et al. [28] theoretically discussed the importance of a collaborative network to improve sustainability in HCSCs. This will aid practitioners and policymakers with investment insights on a range of collaborative networks and practices.

Considering the perspectives of analytical and mathematical approaches in sustainable HCSCs, Hossain et al. [29] recognized and evaluates significant factors of social sustainability in HCSCs by using the Bayesian network approach. The proposed framework will determine potential factors which affect the social sustainability of HCSCs. Maghsoudi et al. [28] investigate the impact of severe disruptions on HCSCs performance by analyzing those disruptions and simulating several proactive and reactive mitigation strategies to ensure the highest possible service level. The findings of this study help to prioritize the most and least significant mitigation strategies to start work with.

There has been a plethora of studies considering the sustainability of HCSCs using MCDM approaches. Elabed et al. [30] proposes a sustainability-oriented innovation (SOI) driven assessment guide and decision-making framework for healthcare managers to enhance sustainability. The selected SOI criteria have been assessed and prioritized using several MCDM approaches such as the Best–Worst Method (BWM), Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), and Fuzzy Best–Worst method (FBWM). This study concludes by emphasizing knowledge of SOI criteria to establish sustainable HCSCs. Moreover, Aljaberi et al. [31] established a framework for assessing and prioritizing sustainability factors and subfactors in HCSCs. This study uses the Analytical hierarchical process (AHP) methodology to develop the hierarchy of sustainability factors. This will provide thoughtful insights to measure and implement sustainability in HCSCs. Considering social sustainability barriers, Ahamat [32] develops a framework for the identification and prioritization of those barriers in HCSCs. The analytical hierarchical process (AHP) is used to prioritize the prime categories and sub-categories of the barriers. Finally, "lack of management support", "lack of commitment" and "lack of coordination" were found to be the top relevant barriers to HCSCs. Leksono et al. [33] developed a sustainable healthcare supply chain performance measurement (SHSCPM) model using a balanced scorecard (BSC) with a decision-making trial and evaluation laboratory (DEMATEL) and analytical network process (ANP). This study finds out the most influential performance indicators which will assist managers and policymakers of HCSCs to give more attention to a particular one over others. However, Khosravi and Izbirak [34] demonstrates a stochastic exponential distribution model for measuring the social sustainability status of a supply chain based on stakeholder theory. The resulting sustainability index can be used to measure the sustainability level of HCSCs.

2.3. Cybersecurity in HCSCs

The importance of cybersecurity in HCSCs is gradually increasing because of their utmost significance and necessity to protect healthcare information systems. Furlanetto et al. [35] analyzed how decisions related to information security practices in HCSCs contribute to protecting patient information. The findings indicate along with being acquainted with the critical information and internal processes; it is necessary to establish necessary metrics to measure the impacts of possible failures. In another study, Kelli et al. [36] present a multi-layer tool framework that provides a particularly successful secure system designed to address eHealth requirements, as well as a blockchain access control component for providing distributed access control to records and data. Furthermore, blockchain may offer a high level of privacy protection. Lee [37] proposed a cyber risk management framework that discusses a cyber risk assessment process for continual improvement of cybersecurity performance. Along the same line, Kshetri [38] goes through the importance of blockchain in identifying the sources of risk in supply chains using Internet of Things (IoT) devices.

2.4. Need for a compressive index for resilience, sustainability, and cybersecurity measures

In the supply chain context, there has been a growing focus on establishing integrated measurement frameworks for resilience, sustainability, and cyber security. There exists a need to establish an assessment framework of resilience, sustainability and cybersecurity. As a numerical value of a supply chain's condition can be established and

used as a tool for more rigorous analysis on the supply chain. For instance, Ghasemi et al. [39], Ghasemi et al. [40], Choukolaei et al. [41], Modibbo et al. [42], Hosseini et al. [43] and Choukolaei et al. [44] demonstrated the assessment of supply chain measures in their studies involving problems such as designing of a blood supply chain network minimizing total cost and maximizing total utility, decision making on location-routing-inventory problems in humanitarian relief chain, selection and analysis of relief centers in a disaster response supply chain, supplier selection problem in a pharmaceutical supply chain, performance evaluation of emergency centers in disaster response, evaluating the efficiency of emergency centers in disaster response. Within the same goal, Jamali et al. [45] introduced a game theoretic approach to obtain a resilience index in the context of renewable energy producers. Yuen Lo et al. [46] had to develop a sustainability index to succeed in their contingency analysis of microalgae supply chains. Along the same vein, El Amrani et al. [47] also proposed Supply chain sustainability index (SCSI) for biomass supply chain. Sardesai and Klingebiel [48] showed an intention to develop a comprehensive index for supply chains however did not capture the cybersecurity context. In another study, Rahman et al. [49] also proposed a viability index for green supply chains. Integrated SCOR 4.0 and machine learning approaches for resilient and sustainable supplier selection have been explored by Khan et al. [50]. Another stream of research made considerable efforts to identify cybersecurity assessments in the supply chain. Works by Álvarez et al. [51] and Shukla et al. [52] are a few examples of that. On the other hand, Cloramidine and Badaruddin [53] came up with a Global Cybersecurity Index (GCI) but it lacks supply chain measures. Thus, it is evident that there is real interest in combining different supply chain measures. Based on the discussion above, however, it appears that these studies either developed standalone supply chain network indices or combined at most two measures. Moreover, cyber security issues were not considered or were only considered in a very limited way in the context of supply chain management.

In sum, although there is a growing amount of recent interest in an integrated index of supply chain measures, the thread of discussion presented above clearly reveals a significant existing gap in the pertaining literature—no comprehensive index is available that considers all these three measures at the same time. Further, there is a lack of frameworks that have been developed to consider the uncertainty to assess the overall comprehensive measure, especially for healthcare supply chains. To fill this void, to bridge that gap, we proposed a novel comprehensive measure called "cyber-susiliency" that combines three key measures (Resilience, sustainability, and cyber-security)of supply chains. To that end, we adopted a data fusion technique called Dempster Shafer Theory. Dempster Shafer's theory combines subjective and uncertain data to get complete overall data. The contribution of this study is highlighted below

- Identify the key factors that contribute to "cyber susiliency" of healthcare supply chain.
- Introduction of a new measure called "cyber-Susiliency" that combines resilience, sustainability, and cyber-security of supply chain.
- Provides holistic framework to assess a comprehensive index for "Cyber-Susiliency" under uncertainty.
- Demonstrate efficacy of hierarchical evidential reasoning in handling healthcare supply chain issues.

3. Methodology

This section includes the theoretical explanation of different analytical approaches that have been used in this study, followed by the framework of this study and factor descriptions. Three independent decision-making methods are employed to construct the framework for assessing the cyber-susiliency competency of a healthcare supply chain: AHP, Dempster–Shafer (DS) theory, and Yager's recursive rule of combination. The study's motivation is to solve the issue of measuring HCSC cyber susiliency competency, which necessitates information from multiple sources throughout the HCSC and organization. The suggested method uses AHP and DS theory to derive optimal attribute weights and fuse data from various sources to develop a quantitative index of cyber-susiliency competency. Yager's rule is used for validation. Different phases of the methodology for assessing cyber susiliency is depicted in Fig. 1.

To identify factors or attributes of cyber susiliency of HCSC, a detailed and thorough review of the relevant literature is conducted in the preliminary phase. The 'Web of Science' platform was used to conduct an extensive search for relevant papers for this purpose. We searched the database for peer-reviewed papers, book chapters, and proceedings in order to gain a broader view of all aspects of HCSC resilience. Over 250 papers were found during the initial search and screening process. Then, the pool of papers to be reviewed was then narrowed by a more thorough screening process based on relative keywords. We further screened the papers by reviewing their abstracts to determine their applicability and suitability. Out of those, seventeen papers related to only the cyber-susiliency (discussed all these aspects same time) of HCSC were considered to extract the salient attributes/measures. These attributes were evaluated by experts from organizations in the relevant fields. These factors or attributes were used to develop a hierarchical decision tree with upper and lower levels, as shown in Fig. 2. In this assessment, the upper level represents the goal, which is to establish a cyber-susiliency index of HCSC, and the lower level represents the attributes and sub attributes of cyber-susiliency.

3.1. Analytical Hierarchical Process (AHP)

Analytical Hierarchical Process (AHP), which Thomas L. Saaty has proposed, became one of the most predominant approaches used for decision-making problems [54]. The foundation of the AHP study is the structure of the problem, with the overall objective at the top level and the specific criteria arranged in the subsequent level [55]. A ranking of all criteria is achieved by using a pairwise comparison system of the hierarchy elements. This technique (AHP), which is based on physical evidence, experience, and intuition, has the advantage of being able to address both qualitative and quantitative components of a decision-making problem [56]. Its usability, effortlessly reasonable system, and easily separable into smaller steps act as distinct benefits over other decision-making approaches [57]. In this study, AHP has been used in weight selection by employing ambiguous and subjective data. The AHP approach is used to rank and assign weights to the criteria that are chosen for the assessment of the resilience, sustainability, and cybersecurity of the Healthcare supply chain. This method's computation comprises a pairwise comparison of the components according to their pertinent relevance, which yields a numerical weight measure for every factor in relation to each other.

The following section shows the action steps involved in the AHP approach:

1. The AHP procedure starts by determining the problem's goals and objectives.

2. After establishing the goal, the hierarchy is structured starting with the highest level (objectives), moving down to the next level (criteria), and finishing with the lowest level (alternatives).

3. The AHP approach employs pairwise comparison to generate a (n x n) matrix for each element in the tier or level immediately above, as well as one matrix for each lower tier. Based on the relative priority of each element over every other element, pairwise comparisons are done.

4. There must be an n(n−1)/2 number of comparison decisions for each matrix in step 3.

5. The consistency index (CI), which has the formula $CI = {}_{max} -n)/(n-1)$ and has n as the matrix size, is calculated in this stage using the eigenvalue $_{max}$ to assess consistency. The consistency ratio's value indicates whether or not the judgment is reliable. If the consistency ratio is f0.1, the judgment matrix is acceptable. When judgments are contradictory, the judgment has to be strengthened.

6. The eigenvectors are weighted using hierarchical synthesis after being normalized by the weights of the criterion.

Fig. 1. Phases of the methodology for assessing cyber susiliency.

3.2. Dempster–Shafer (DS) Theory

The Dempster–Shafer (DS) theory addresses uncertainty through belief functions [58], also referred to as evidence theory, which can be harnessed to solve the problem. It is a data fusion technique initiated by Arthur Dempster [58] and afterward expanded by Glenn Shafer in 1976. DS theory is a well-established formalism for reasoning and making decisions with uncertainty [59]. It is based on representing independent pieces of evidence by completely monotone capacities and combining them using a generic operator called Dempster's rule [60]. When compared to other methods, the Dempster–Shafer theory offers a significant edge in terms of reasoning with uncertainty and subjective data effectively and efficiently. Due to its provision of accurately allocating evidence to not only recognize the disjoint but also the non-disjoint sets, DS theory, in comparison to Bayesian theory, enables one to have a clearer expression of uncertainty information [61]. Another noteworthy aspect of the DS theory is that it offers the possibility of giving focal components and discerning frame probabilities to be assigned [62].

The frame of discernment ( ), according to Dempster–Shafer's theory, is a collection of mutually incompatible hypotheses that may be described as follows: = a, b, c, taking into account three assumptions. The power set, indicated by 2 , is the set containing all subsets of ( ), and in our case, it consists of the following components.

$$2 = \{\emptyset; \{a\}; \{b\}; \{c\}; \{a; b\}; \{a; c\}; \{b; c\}; \{a; b; c\}\} \quad (1)$$

Every component of this power set is referred to as a focus element; the a, b, and c components signify ignorance because they offer no precise details. This focus element can be given a belief level ranging from 0 to 1, with 0 denoting no belief and 1 denoting total belief. A basic probability assignment (BPA), also known as a mass function (m) such that $0 f\ m(A)\ f\ 1, \forall A\ ; m(\emptyset) = 0$ denotes the level of belief for each hypothesis [61]. According to Eq. (1), the statement or hypothesis